

Product Requirements Document (PRD): CampusGuard

Version: 2.0 | **Developer:** asras.ai | **Status:** Draft for Development

1. Project Vision

CampusGuard is an enterprise-grade, multi-site Visitor Management System (VMS). It replaces traditional handwritten notebooks with a high-speed, AI-enhanced digital workflow designed for complex industrial and corporate environments. The system ensures "Zero-Wait" for VIP customers and strict accountability for employees.

2. Technical Architecture & Stack

- **Style:** Microservices Architecture (FastAPI)
- **Language:** Python 3.x
- **Web Framework:** FastAPI (High performance, async support)
- **Database:** PostgreSQL (Relational master data & logs)
- **Storage:** MinIO (Object storage for live visitor photos)
- **Messaging:** Kafka & Zookeeper (Async notifications & background tasks)
- **Deployment:** Docker & Docker Compose (Containerized for multi-site edge/cloud)

3. Master Data & Settings Setup

3.1 Site & Operations

- **Site Master:** ID, Name (e.g., Plant A), Location, Timezone.

- **Department Master:** Name, Department Head Name, Email ID (Primary target for automated reporting).
- **Building/Area Master:** Specific zones (e.g., Server Room, Production Line).
- **Gate Master:** Specific entry/exit points (e.g., Main Gate, VIP Gate).

3.2 Visitor Types & Branding

- **Customer:** **RED BANNER** - VIP path, skip-able photo capture.
- **Vendor/Contractor:** **BLUE BANNER** - Supports multi-entry/long-term passes.
- **General Guest:** **GREEN BANNER** - Standard entry workflow.
- **Temp Employee:** **YELLOW BANNER** - For forgotten ID tracking.

3.3 Notification & Global Settings

- **Notification Config:** SMTP (Email), SMS Gateway, WhatsApp Business API.
- **UI Customization:** Company Logo, Themes (Dark/Light), Multilingual Support (English, Tamil, etc.).

4. Core Functional Modules

4.1 Responsive Gate Interface (Desktop & Tablet)

- **Single Web Platform:** Fully responsive. Tablet view optimized for touch; Desktop view for peripheral management (A6 printers/webcams).
- **Visitor Request (Pre-Registration):** PAs/Admin raise requests; visitors receive a QR/Link via WhatsApp/SMS for instant gate check-in.

4.2 Entry & Exit Workflow

- **Visitor Info:** Name, Phone, "Coming From" (Org/City), Purpose, Meet With (Staff search).

- **Identity Capture:** Input ID Type and ID Number (Manual entry; no document upload).
- **Asset/Accessory Tracking:** Log items (Laptops, Tools) with **Serial Numbers**.
- **Logistics:** Vehicle Type and Registration Number capture.
- **Live Photo:** Capture via webcam/tablet; skip-able if photo is pre-uploaded by Customer.

4.3 AI Features (Core & Roadmap)

- **Sentiment Analysis:** AI parses "Purpose of Visit" to flag aggressive/suspicious intent.
- **Intent-Based Routing:** Auto-notifies additional stakeholders (e.g., Maintenance) based on text keywords.
- **Predictive Overstay:** Machine learning predicts visit duration and alerts security to anomalies.
- **Fuzzy Blacklist Matching:** Matches names/phones even with slight spelling variations.
- **Roadmap:** OCR for ID cards, ALPR for vehicle gates, and Face Recognition check-in.

5. Security & Reporting

- **Forgotten ID Protocol:** Security issues Yellow badge; system auto-emails **Department Head** and Security Manager.
- **Dashboard Views:** - **Global:** Company-wide stats, cross-site blacklists, and audit logs. - **Local:** Site-specific live visitor feed and overstay alerts.
- **Anti-Passback:** Prevents duplicate entry flags for active sessions across different gates.